



GOING GREEN IS A BILLION DOLLAR EFFORT

At the World Economic Forum held at Davos from 16th-20th January 2023, among the key things of discussion emerged the broader theme of sustainability, green energy and green mobility for a zero-carbon future. Dr Prateek Kanakia, Chairman and Founder of TheGreenBillions Limited (A company specialising in consulting, managing, and executing sustainable initiatives), shares his views on sustaining the transition from conventional fuel to green energy through the provision of 'Green finance', clarity of policy matters, hand holding players in the renewable and alternative energy sector and extracting green hydrogen from city's solid waste.

What are your thoughts about the relevance of alternative energy, and going green for Indian businesses?

Sustainability is the mantra for way forward, even as use of conventional energy to produce a new form of energy, which is less in quantum, high on prices today, is being debated. Recently at World Economic Forum, an initiative called 'Giving to Earth Action' was floated, which suggests allocation of 3 Trillion USD, called as 'Climate finance'. The need for 'Green finance' was raised by the Government of India at COPT 27, where these kinds of financial hand-holdings by governments, institutions and authorities could address viability gaps for attaining sustainability as a concept. When it comes to adopting and consuming for example green hydrogen the response of industry and corporates, is that the cost matrix is not viable because they work for bottom lines, top lines. If you want to become a hub or the narrative which is being set by the Government of India about exporting Green Hydrogen those economy of scale will have to be reached through a lot of sustainable financing. The GreenBillions Limited is only one such entity in the country, which is taking an initiative to extract hydrogen from municipal solid waste, and to this effect have signed a Public Private Partnership (PPP) with Pune Municipal Corporation to treat 350 tonnes of municipal solid waste for next 30 years, and likely to achieve 10 tonnes of hydrogen per day.

What are the hindrances to these transitions of energy modes?

There is difficulty involved in going green because it impacts

the stakeholder's current business model, balance sheet. When solar, wind alternative fuels were introduced they were expensive, but after two decades at least our country has been able to achieve a little more cost efficiency. There is still a stark gap in these new alternative energy, where the process from start to delivery costs a bomb e.g., Green hydrogen. Though a mandate by the government to blend green hydrogen back to the city as distribution exists, there are no accompanying safety guidelines as of today. A lot of lacuna exists in understanding how to take the transition even if the company is debt free, is showing a lot of equity and surplus to invest into energy transition. Although, the first draft of the National Green Hydrogen Mission has been released by the Government of India, many clarifications are still needed. The Ministry of New and Renewable Energy considers waste to energy as renewable and Ministry of Petroleum and Natural Gas considers any hydrogen achieved and generated out of renewables as green. If these two definitions are integrated, we still remain clueless about our status of being green or not. We spoke to a few PSUs, but they still do not have clarity on how to blend it back because every pipeline is a different pipeline in this country and this grid was designed, when hydrogen was not into play. Besides clarifications from the government, there has to be a lot of hand holding to be done to overcome this situation.

How large is the Pune Municipal solid waste opportunity and the need to look at plans like the ones you are working on?

When it comes to impact it can't be bigger and better than

The GreenBilion as it involves treating the city solid waste, converting it to a superlative fuel like hydrogen and blending it back to the city gas distribution, firing industries. As municipal solid waste is 25 per cent methane, it is more detrimental, hazardous and harmful to the environment, while electrolyser as a process is energy negative where more energy is consumed to produce less energy, and we end up paying more for green hydrogen as it is green energy. The situation is ironical where water conservation is the aim, but it is used as raw material for producing fuels. Also, biomass is extremely necessary for replenishing the soil, which is a natural process for maintaining ecological balance. When it comes to the most impactful space i.e., municipal solid waste of any city, from it, I want to establish the cleanest form of hydrogen. People must also know that we are using technologies developed and invented by premier institutions like Bhabha Atomic Research Centre and Indian Institute of Science and are not being borrowed from anywhere else from the globe. Interestingly, very project of mine would cater to four major national ambitions, namely, Swachh Bharat, Atal Mission on Urban Rejuvenation and Transformation, National Green Hydrogen Mission, and Atma Nirbhar Bharat. My idea is to replicate this project of this scale and nature across the country because finding land for landfills involves costs and building infrastructure for landfills in villages on city outskirts has led to agitations. Land will not expand, but the population is increasing, so to have newer landfills is a pain point, and the kind of premium that real estate attracts, landfills will dent healthy revenue of the government.